AO RHYTHM · DIGITAL TWIN CHALLENGE 2026

How our heart twin spots fluid buildup early

After heart treatment, many patients go home to towns far from a heart specialist. Fluid can quietly build up in the lungs. Our digital twin watches for it and warns the doctor early.

STEP 1

Patient wears a chest patch

At home, a small patch records the heartbeat, breathing, body position and skin temperature.

STEP 2

The recording is checked

The computer first makes sure the signal is clear. If it is too noisy, no guess is made.

STEP 3

AI estimates the lung pressure

An AI model reads the signals and estimates the pressure in the lungs' blood vessels. High pressure means fluid is building up.

STEP 4

A clear answer

- Fluid building up
- Not sure yet: doctor checks
- No fluid buildup

STEP 5

Doctor sees who needs help first

On the doctor's screen, patients most at risk are at the top of the list, even from a city far away.

STEP 6 · THE DIGITAL TWIN

A 3D heart copies the patient's heart

On screen, a 3D heart beats in time with the patient's real heartbeat, and glows red when fluid is building up.

taught the AI

HOW THE AI LEARNED

From real hospital patients

It studied recordings from 59 heart-failure patients whose true lung pressure was measured with a thin tube inside the heart.

shown on screen

SUGGESTIONS FOR THE DOCTOR

Based on medical guidelines

Suggestions follow published heart-failure guidelines. If the system is not confident, it stays quiet. It never suggests a dose.

A student prototype for learning, tested on recorded data (PhysioNet). Not a medical device. The doctor always decides.